Sachin Bala

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Technical Skills

Python (Pandas, NumPy, scikit-learn, statsmodels), SQL (MySQL), R, Machine Learning, Anomaly Detection, Statistical Modeling, Causal Inference, A/B Experimentation, DOE, SPC, Process Capability Studies, Data Visualization (Grafana, Matplotlib), Computer Vision (YOLO), JMP, Minitab, LabVIEW, SolidWorks, Six Sigma, PFMEA, Lean Manufacturing, Microsoft Suite, Cognex, Dr Schenk vision system

Education

Western Michigan University (WMU) **Overall GPA: 3.69**

Bachelor's in Mechanical Engineering with minor in Computer Science and Mathematics

Work Experiences

Associate Process Development Engineer, Tesla Inc., Austin, TX **02/25-present**

- Developed and deployed statistical models and anomaly detection pipelines using Python, R, MySQL, and JMP to identify yield excursions and tool failures in high-volume manufacturing; automated alerting workflows that surfaced issues before they impacted production targets.
- Translated ambiguous operational problems into structured data science questions using DOE, SPC, FMEA, and capability analysis; delivered causal findings and actionable recommendations to engineering and business stakeholders.
- Commissioned two electrode production lines achieving 98% yield by combining rigorous process experimentation with real-time metric tracking, enabling data-driven investment and operational decisions across the manufacturing organization.

Engineering Technician, Process Development Team, Tesla Inc., Austin, TX **8/23-02/25**

- Built a real-time anomaly detection pipeline in Grafana monitoring tool vibration metrics across the production floor; automated threshold-based alerts reduced unplanned failures and enabled proactive corrective action.
- Applied statistical capability studies and structured A/B process experiments to validate tools against product specifications, identifying improvements that increased production rate by 10% while sustaining quality targets.
- Performed root cause analysis on downtime events using SPC charts, fault logs, and process historians; quantified impact and implemented corrective actions that improved operational reliability and uptime.

Engineering Co-Op, Scherdel SST, Muskegon, MI **8/22-12/22**

- Analyzed EV component market and manufacturing material data to support product launch decisions; synthesized quantitative findings for cross-functional stakeholders.

Process Engineering Intern, Scherdel SST, Muskegon, MI **05/22-08/22**

- Built a predictive failure-detection model using ML algorithms and statistical analysis, improving manufacturing efficiency and accuracy by 25%; validated using Gauge R&R methodology and F1-score metrics.
- Developed a Computer Vision defect detection system; used causal validation methodology (Gauge R&R, F1-score analysis) to ensure model reliability.

Laser Tech Technician Internship, Technique Inc., Jackson, MI **09/21-04/22**

- Optimized laser cutting using CAD and machine troubleshooting; validated chassis parts via design review and laser tracker measurement data.
- Updated SOPs and trained technicians and operators to follow standardized procedures.

Engineering Research and Project

Senior Design Project - Two Tier Industrial Cart (Technique Inc.) **01/22-12/22**

- Designed a two-tier industrial cart in SolidWorks with scissor lift mechanism; validated with FEA analysis and numerical calculations.
- Applied DFMEA to identify and mitigate risk, establishing data-backed design controls and a quality control plan.

Honors and Membership

Diether H. Haenicke Scholarship, WMU Dean's List, Yellow Belt Training (Six Sigma), Root Nepal (Co-Founder)